

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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Register Number	192424214

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 50

Code of Conduct

I M.Kasif Jala certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.Kasif Jalal

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

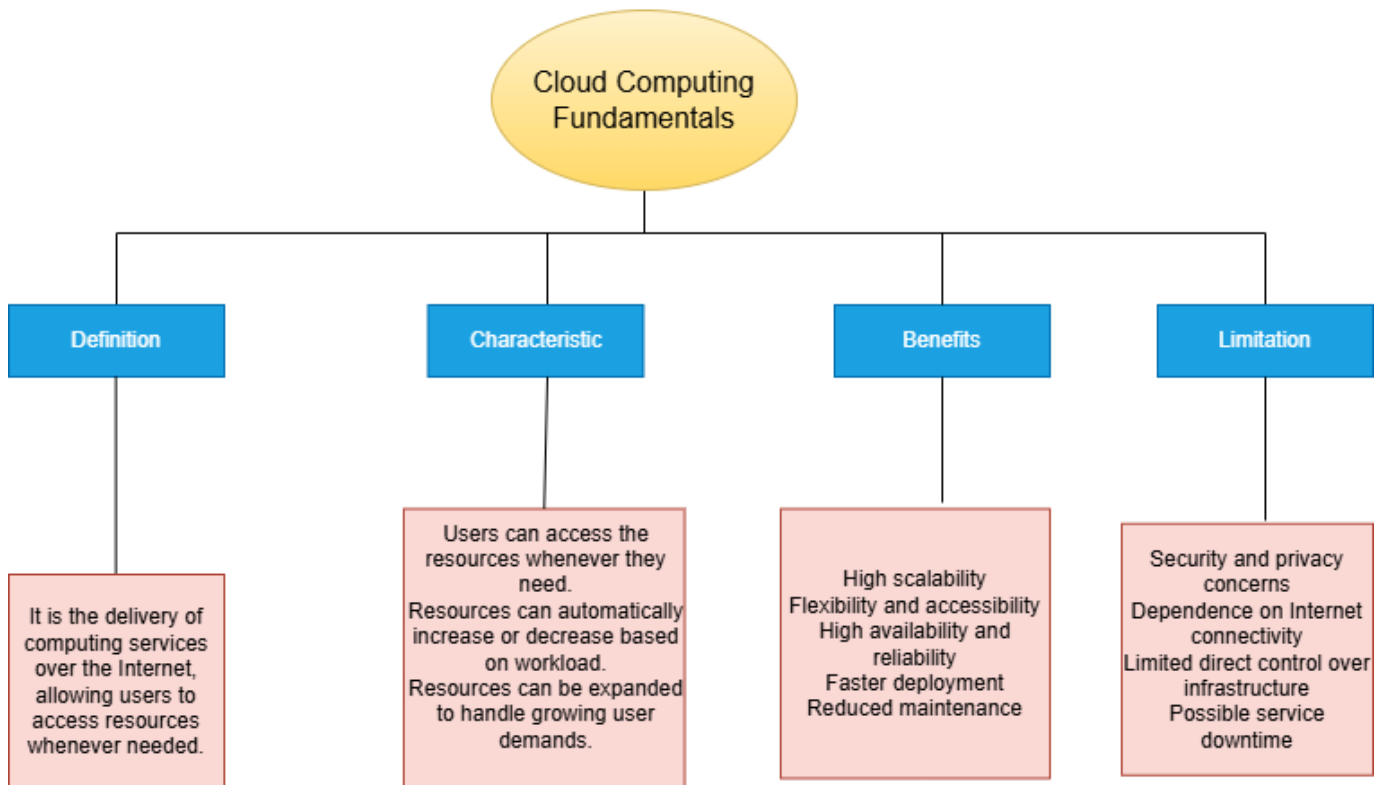
Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

5. Questions

i). Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?
ii). Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?
iii). Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?
iv). Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?
v). Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

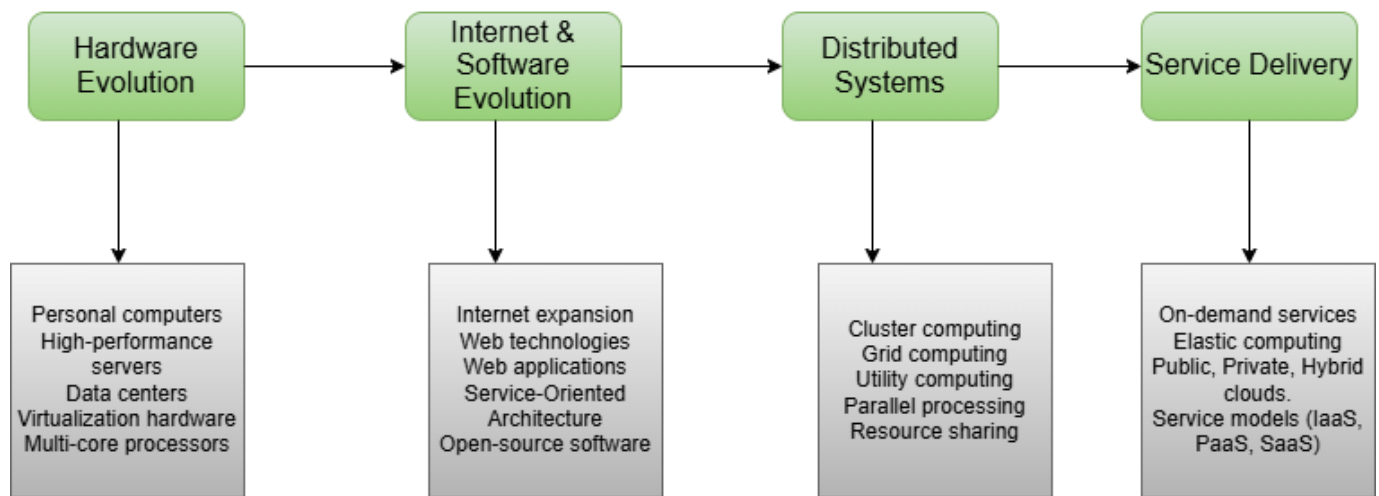
1, Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.

Concept Map 1: Cloud Computing Fundamentals



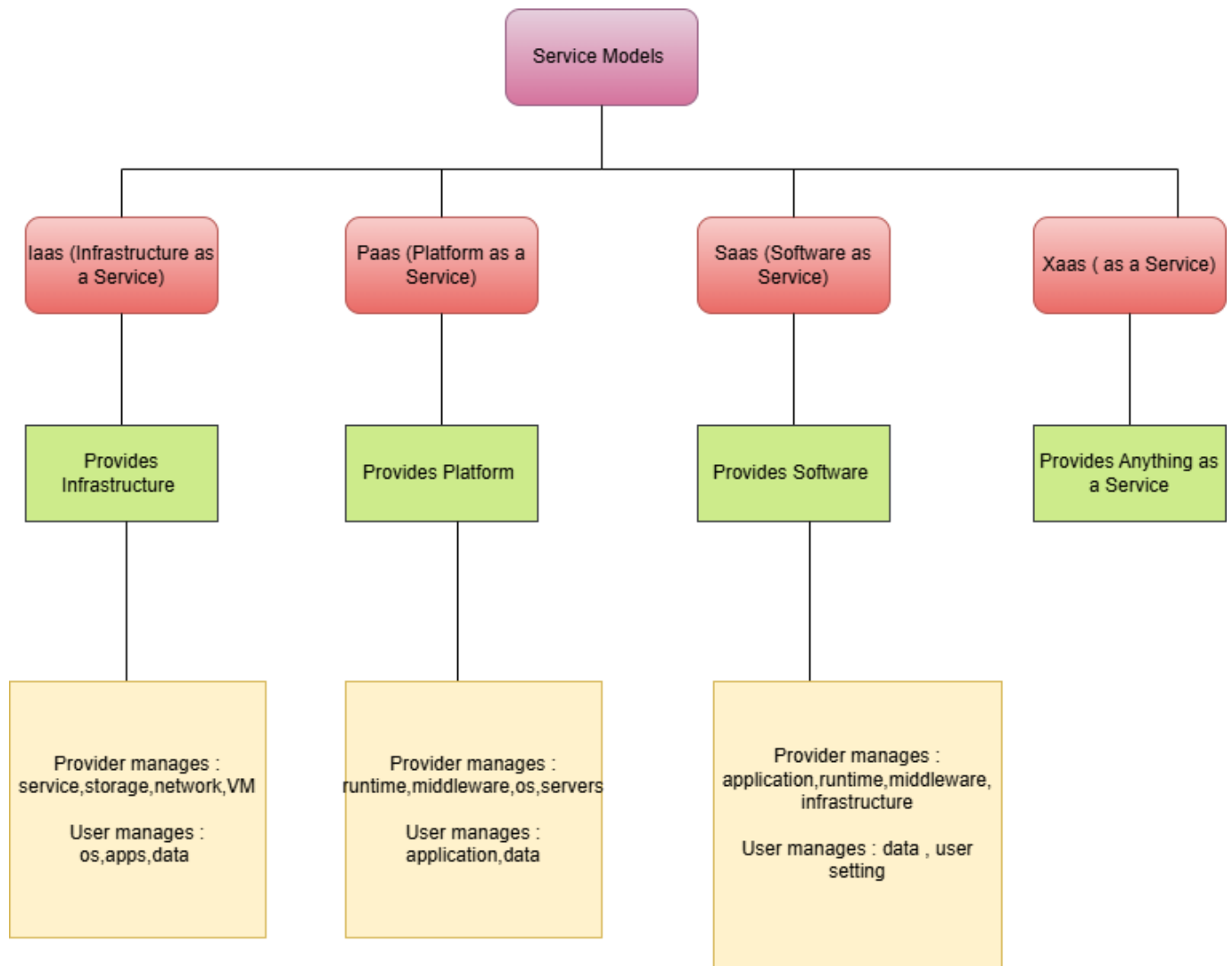
2, Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.

Concept Map 2: Evolution of Cloud Computing



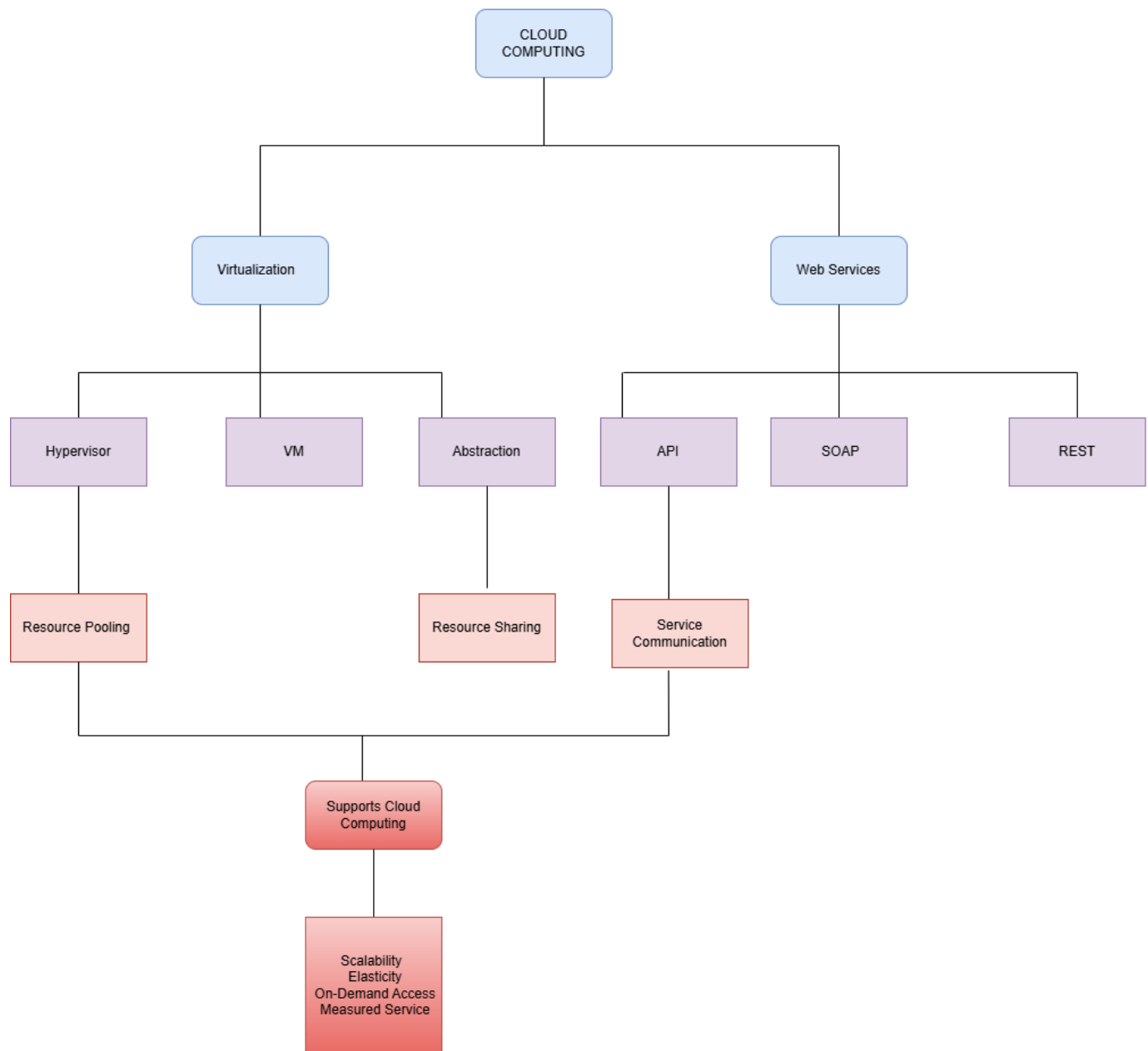
3, Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.

3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS



4, Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Concept Map 4: Virtualization and Web Services as Enablers of Cloud



i). Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?

The concept map effectively represents cloud computing principles by organizing the main concepts in a logical hierarchy, starting from cloud computing and branching into service models, virtualization, web services, characteristics, and benefits. The use of connected nodes clearly shows the relationship between concepts, making the architecture easy to understand. The decision to separate provider responsibilities from user responsibilities in the service model and to show virtualization and web services as the key enablers of cloud computing improves clarity. This structure helps learners understand how different cloud components interact to deliver scalable and efficient services.

ii). Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?

Scalability allows cloud systems to increase or decrease resources according to user demand, while elasticity automatically adjusts resources during workload changes. Resource management ensures efficient allocation and utilization of computing resources. Together, these features improve system performance, availability, and cost efficiency. Without scalability and elasticity, the system could become overloaded during peak demand or waste resources during low demand. Poor resource management would result in inefficient utilization, increased operational costs, reduced performance, and poor user experience.

iii). Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?

The concept map includes APIs, SOAP, REST, virtualization, and cloud service models because they are fundamental components of modern cloud computing. APIs allow applications to communicate with cloud services, while REST and SOAP provide standardized methods for data exchange between systems. Service models such as IaaS, PaaS, and SaaS demonstrate different levels of cloud service delivery. These technologies improve interoperability, enable seamless communication, simplify application development, and enhance overall system efficiency by providing flexible and reusable cloud services.

iv). Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?

The relationships among virtualization, web services, service models, and cloud infrastructure demonstrate how cloud computing operates as an integrated system. Virtualization enables resource sharing and efficient hardware utilization, while web services provide communication between applications. Service models deliver computing resources based on user requirements. These interactions improve reliability through redundancy and high availability, enhance performance by efficiently allocating resources, and provide a better user experience through on-demand access, faster deployment, and scalable services.

v). Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

Developing the concept map improved my understanding of cloud computing by showing how different technologies and service models are connected within a single architecture. It helped me understand the roles of virtualization, APIs, web services, scalability, and resource management in delivering cloud services. If I were redesigning the concept map, I would include deployment models such as Public Cloud, Private Cloud, Hybrid Cloud, and Community Cloud, along with practical examples of cloud providers like AWS, Microsoft Azure, and Google Cloud Platform. These additions would make the concept map more comprehensive and easier to relate to real-world cloud computing environments.